6ixtype Ecosystem + 6Data Full Product Build Plan

0. Ecosystem Definition

The ecosystem must only contain these core products/areas:

```
6ixtype = parent brand / portfolio / launchpad
6Stats = Spotify/music analytics
6Data = main data platform
6Play = music/audio software direction
Research/Learning tools = inside 6Data
```

Do not create or reference extra branded products outside this list.

Do not introduce separate product names such as operations products, finance products, fitness products, or other 6-branded side apps.

If additional ideas are needed, they must be expressed as **modules, templates, dashboards, connectors, or modes inside 6Data**, not as separate brands.

Correct structure:

```
6ixtype
├── Launchpad / portfolio / parent brand
├── 6Stats
│   └── Spotify/music analytics app
├── 6Data
│   ├── Import Centre
│   ├── Data Vault
│   ├── Dataset Profiles
│   ├── Cleaning Studio
│   ├── Query Lab
│   ├── Visualisation Studio
│   ├── 3D Lab
│   ├── Render Studio
│   ├── Research Workspace
│   ├── Learning Mode
│   ├── Dashboard Builder
│   ├── Automation Centre
│   ├── Publishing Hub
│   └── Connector Marketplace
└── 6Play
    └── Music/audio software direction
```

---

1. Overall Product Philosophy

The goal is not to build a basic portfolio with a few projects.

The goal is to build a real software ecosystem under **6ixtype**, where:

- **6ixtype** acts as the public identity, launchpad, product hub, and portfolio.
- **6Stats** proves the ability to ship a real deployed analytics product.
- **6Data** becomes the flagship platform: a robust data operating system.
- **6Play** remains the music/audio software direction.
- Research and Learning tools live inside **6Data**, because they are data workflows.

The main strategic idea:

```
6Stats proves music analytics.
6Data generalises analytics into a full data platform.
6ixtype presents the ecosystem.
6Play continues the music/audio direction.
```

6Data should not feel like a small feature. It should feel like a future-forward data environment.

---

2. 6Data Product Vision

Main idea

6Data is a full data operating system where users can bring in data, understand it, clean it, query it, visualise it, render it, research it, learn from it, automate it, and publish it.

Simple pitch:

```
6Data turns raw data into understanding.
```

Longer pitch:

```
6Data is a personal data lab for importing files, databases, APIs and app
data, then profiling, cleaning, querying, visualising, researching, learning
from and publishing the results.
```

Even longer product definition:

```
6Data is a browser-based data workspace that combines the useful parts of
spreadsheet tools, SQL editors, dashboard builders, research notebooks,
```

```
learning environments, and 3D visualisation systems into one practical
platform.
```

---

3. What 6Data Must Be Able To Do Eventually

6Data should eventually support the full data lifecycle:

```
Source
→ Ingest
→ Profile
→ Clean
→ Transform
→ Join
→ Model
→ Query
→ Visualise
→ Explore in 3D
→ Render
→ Explain
→ Research
→ Learn
→ Dashboard
→ Publish
→ Export
→ Automate
→ Reuse
```

The app should be designed so each part of the lifecycle is traceable.

A user should always be able to answer:

```
Where did this data come from?
What changed?
Who changed it?
What cleaning steps were applied?
Which query produced this chart?
Which dataset version was used?
Can I reproduce this result?
Can I export it?
Can I share it?
Can I learn from it?
```

---

4. Main User Types

6Data should be useful for multiple types of users.

1. Normal personal analytics user

Wants to upload or connect data and understand patterns.

Examples:

```
Spotify listening history
spending CSV
sleep tracker export
game stats
habit tracker
personal survey
```

2. Student / learner

Wants to learn:

```
SQL
data cleaning
statistics
charts
databases
research methodology
data visualisation
basic data science
```

The app should teach using the user's own data where possible.

3. Research user

Wants to create a reproducible research workflow:

```
research question
hypothesis
dataset upload
data cleaning
analysis
charts
findings
```

```
limitations
report export
```

4. Developer / technical user

Wants to connect:

```
Postgres
Supabase
SQLite
JSON APIs
CSV exports
app data
```

They need SQL, schema explorer, saved queries, and export tools.

5. Creator / portfolio user

Wants to create impressive visual outputs:

```
dashboards
3D views
data renders
data sculptures
interactive visuals
report graphics
shareable snapshots
```

---

5. 6ixtype Role

6ixtype is the parent brand.

It should include:

```
Home
Launchpad
Projects
About
Contact
```

The important page is **Launchpad**.

Launchpad purpose

Launchpad is the product ecosystem page.

It should show only the official core areas:

```
6Stats
6Data
6Play
6ixtype
```

Research/Learning should not be a separate product card. It should appear as a capability inside the 6Data card or 6Data detail page.

Launchpad cards

6Stats

```
Status: Live
Type: Music analytics
Description: Spotify listening analytics, top tracks, top artists, listening
history, dashboards and music-data insights.
Primary action: Open app
Secondary action: View case study
```

6Data

```
Status: Building
Type: Data platform
Description: Import, clean, query, visualise, research, learn from, render
and publish datasets from one workspace.
Primary action: View product
Secondary action: View roadmap
```

6Play

```
Status: Prototype / Direction
Type: Music/audio software
Description: The music and audio software direction under 6ixtype, focused on
playback, creative tools and audio-related experiments.
Primary action: View concept
Secondary action: Coming soon
```

6ixtype

```
Status: Live
Type: Parent brand / portfolio / launchpad
Description: The parent software identity and launchpad for deployed apps,
experiments and long-term products.
Primary action: Open site
```

Do not include

Do not include separate cards for unrelated product brands.

If old projects need to be shown, they belong in the normal **Projects** page, not the Launchpad.

---

6. 6Data Main Navigation

Inside 6Data, use this sidebar:

```
Overview
Workspaces
Import Centre
Data Vault
Datasets
Query Lab
Visualise
3D Lab
Render Studio
Dashboards
Research
Learning
Automations
Connectors
Publishing
Settings
```

Page responsibilities

Overview

The main control panel.

Shows:

```
recent workspaces
recent datasets
```

```
recent dashboards
recent research projects
recent imports
quick actions
demo datasets
data quality warnings
```

Workspaces

Organises projects and datasets.

Example workspaces:

```
Spotify Research
University Research Project
Personal Analytics
Demo Data Lab
Music Data Exploration
```

Import Centre

Where data enters the app.

Sources:

```
CSV
JSON
Excel later
SQLite upload
Postgres later
Supabase later
REST API later
Google Sheets later
6Stats import
manual table
demo dataset
```

Data Vault

Long-term storage area.

Shows:

```
raw files
cleaned versions
dataset versions
exports
```

```
snapshots
attachments
```

Datasets

Main list of datasets.

Each dataset shows:

```
name
source type
rows
columns
quality score
last updated
workspace
status
```

Query Lab

SQL and visual query tools.

Visualise

Normal 2D charts and advanced analytical visuals.

3D Lab

Interactive 3D data exploration.

Render Studio

Polished data-driven renders and visual storytelling.

Dashboards

Saved dashboards made from charts, queries, notes, and 3D snapshots.

Research

Research project builder.

Learning

Learning tools using real datasets.

Automations

Scheduled refreshes, alerts, cleaning pipelines, exports.

Connectors

Database/API/app connections.

Publishing

Share links, exports, reports, snapshots.

Settings

Workspace/user/platform settings.

---

7. 6Data Visual Design

The UI should feel like a serious futuristic lab, not a generic admin dashboard.

Visual style

```
dark mode first
clean cards
glass panels
subtle gradients
data-grid background
soft glow accents
smooth animations
clear typography
developer-tool seriousness
premium dashboard feel
```

Design balance

Avoid making everything too flashy.

The app should have two moods:

```
Analytical mode = clean, readable, precise
Exploration mode = 3D, animated, immersive
```

Charts and tables must stay readable. 3D and renders should feel powerful but optional.

---

8. 6Data Core Data Lifecycle

The product should be built around this lifecycle:

```
1. Import
2. Inspect
3. Profile
4. Clean
5. Transform
6. Query
7. Visualise
8. Explore
9. Render
10. Research
11. Learn
12. Dashboard
13. Publish
14. Automate
```

Each stage should have UI and database support.

---

9. Data Intake Layer

Goal

Let users bring in as much data as possible over time.

MVP sources

Start with:

```
CSV upload
JSON upload
manual table
demo datasets
6Stats demo import
```

Later sources

```
Excel
SQLite
Postgres
Supabase
REST API
Google Sheets
public datasets
6Stats real connector
```

Import flow

```
1. Choose source
2. Upload/connect/paste data
3. Preview sample
4. Detect schema
5. Confirm column types
6. Save raw data
7. Generate profile
8. Create dataset
9. Show next recommended actions
```

Import Centre UI

Sections:

```
Files
Databases
APIs
Apps
Manual
Demo
```

Connector cards:

```
CSV
JSON
Excel
SQLite
Postgres
Supabase
REST API
6Stats
Manual Table
Demo Dataset
```

Each connector card should have:

```
status: Available / Coming Soon
difficulty: Easy / Advanced
type: File / Database / API / App
description
action button
```

---

10. Dataset Understanding Layer

Once data is imported, 6Data should inspect it automatically.

Dataset-level profile

Show:

```
row count
column count
file size
source
import date
last updated
quality score
missing values total
duplicate rows
detected date range
numeric columns
categorical columns
text columns
possible primary keys
possible relationships
warnings
suggested questions
suggested charts
```

Column-level profile

For each column:

```
column name
detected type
final type
nullable
missing count
missing percentage
unique count
top values
sample values
min
max
mean
median
mode
standard deviation
outlier count
```

```
date min
date max
type confidence
```

Quality score

Create a visible quality score using:

```
Completeness
Consistency
Uniqueness
Type confidence
Validity
Freshness
```

Example:

```
Overall quality: 84%

Completeness: 91%
Consistency: 78%
Uniqueness: 96%
Type confidence: 88%
Validity: 81%
Freshness: 100%
```

Warnings examples

```
Column "date" contains mixed formats.
Column "amount" has 12 non-numeric values.
42 duplicate rows detected.
Column "id" looks like a primary key.
Column "artist_id" may join with another table.
Column "duration_ms" has possible outliers.
```

Suggested questions

Generate simple deterministic questions:

```
What are the top categories?
How does this value change over time?
Which rows have missing values?
Which records are duplicates?
What are the highest and lowest values?
```

```
Which columns are most complete?
What patterns exist by day, month or category?
```

---

11. Cleaning Studio

Goal

Let users clean data without destroying raw files.

Principle

Always preserve:

```
raw version
cleaned version
cleaning steps
dataset version history
lineage
```

Cleaning actions MVP

```
rename column
change column type
remove duplicate rows
remove rows with missing values
fill missing values
trim whitespace
replace values
lowercase/uppercase text
filter rows
sort rows
create calculated column
extract date parts
split column
merge columns
```

Later cleaning actions

```
join datasets
union datasets
group categories
detect outliers
remove outliers
```

```
normalise text
standardise dates
validate against rules
create lookup table
deduplicate fuzzy matches
```

Cleaning step example

```
{
  "step_type": "rename_column",
  "description": "Rename played at to played_at",
  "config": {
    "from": "played at",
    "to": "played_at"
  }
}
```

Cleaning Studio UI

Panels:

```
left: cleaning step list
centre: before/after table preview
right: action config panel
top: dataset version selector
bottom: apply/undo/save controls
```

Acceptance requirement

A user should be able to say:

```
I imported this file, cleaned it in these 5 steps, then used the cleaned
version for my chart/report.
```

---

12. Query Lab

Goal

Let users ask questions from data using different skill levels.

Query modes

```
Visual query builder
SQL editor
Saved queries
Query history
Natural language query later
```

SQL editor

Use Monaco-style editor.

Features:

```
write SQL
run query
show result table
show errors
show execution time
save query
duplicate query
turn result into chart
export result
add result to dashboard
attach result to research project
```

Visual query builder

For non-SQL users.

Controls:

```
select dataset
select columns
filter rows
group by
aggregate
sort
limit
preview generated SQL
run query
```

Query result metadata

Each result should know:

```
source dataset
dataset version
query text
run time
row count
generated chart links
created at
```

Natural language query later

Example:

```
Show me total listening time by hour for the last 30 days.
```

6Data should produce:

```
generated SQL
query result
chart suggestion
plain-English explanation
limitations
```

AI must never hide the query. The method must be visible.

---

13. Standard Visualisation Studio

Goal

Let users build normal analytical charts.

Chart types MVP

```
number card
bar chart
line chart
area chart
pie/donut chart
scatter plot
histogram
table
simple heatmap
```

Advanced charts later

```
box plot
calendar heatmap
correlation matrix
pivot table
stacked bar chart
bubble chart
timeline
sankey
network graph
map
```

Chart builder mapping

User chooses:

```
dataset or query result
chart type
x axis
y axis
group/color
size
aggregation
filters
sort
limit
title
description
```

Smart recommendations

The app should suggest visuals based on column types:

```
date + number = line chart
category + count = bar chart
two numeric columns = scatter plot
hour + day = heatmap
single numeric result = number card
category share = donut chart
distribution = histogram
```

Chart traceability

Every chart should show:

```
source dataset
source query
dataset version
filters
last refreshed
```

This matters for trust and research.

---

14. 3D Lab

Goal

Let users explore data spatially and interactively.

3D should not replace normal charts. It should be used for:

```
multi-variable exploration
clustering
outlier discovery
storytelling
music taste mapping
spatial relationships
time evolution
immersive demos
```

3D views MVP

Start with:

```
3D scatter plot
3D bar field
3D terrain/surface
simple 3D timeline
```

3D views later

```
3D network graph
3D geospatial layers
density clouds
cluster galaxies
animated time paths
```

```
data rooms
immersive research scenes
```

3D scatter

Column mappings:

```
X axis
Y axis
Z axis
color
size
label
group
opacity
time
```

Example using music data:

```
X = tempo
Y = energy
Z = valence
color = artist or genre
size = play count
label = track name
```

Interactions

```
orbit
pan
zoom
hover tooltip
click point
filter group
isolate selection
reset camera
save camera preset
export screenshot
add snapshot to dashboard
attach snapshot to research project
```

Performance rules

```
limit rendered points by default
sample large datasets
```

```
use instancing
warn users when data is too large
allow aggregation
avoid freezing browser
```

3D Lab UI

Layout:

```
left: dataset/visual type selector
centre: 3D canvas
right: column mapping panel
bottom: filters/timeline/camera presets
```

---

15. Render Studio

Goal

Create polished, shareable data-driven renders.

Difference:

```
3D Lab = interactive analysis
Render Studio = presentation, storytelling, visuals, art, export
```

Render templates

MVP templates:

```
Data Orb
Category Towers
Timeline Tunnel
Data Terrain
Audio DNA
Network Galaxy later
```

Render configuration

User maps:

```
dataset
columns
```

```
colour mapping
size mapping
height mapping
time mapping
label mode
camera angle
background
style preset
export size
```

Render outputs

MVP:

```
PNG export
dashboard snapshot
research report image
```

Later:

```
video animation
interactive embed
animated GIF
high-res render
```

Render use cases

```
research report cover
dashboard hero
portfolio visual
music taste sculpture
dataset identity image
shareable social visual
presentation graphic
```

Important rule

Render Studio must not invent fake data.

Every render should be linked to:

```
source dataset
mapping config
```

```
render template
created date
```

---

16. Dashboard Builder

Goal

Let users turn data into reusable dashboards.

Dashboard widgets

```
number card
chart
table
query result
markdown note
image/render
3D snapshot
filter control
date range picker
section heading
```

Builder features

```
create dashboard
add widget
move widget
resize widget
remove widget
edit widget
save layout
preview mode
edit mode
duplicate dashboard
export dashboard
share dashboard
```

Dashboard templates

Templates should be generic and inside 6Data:

```
Blank dashboard
Music analytics dashboard
```

```
Research dashboard
Data quality dashboard
Learning dashboard
Personal analytics dashboard
```

Do not create new product brands for templates.

Traceability

Each widget should know:

```
source dataset
query
chart config
last refresh
```

---

17. Research Workspace Inside 6Data

Goal

Make 6Data genuinely useful for research and reproducibility.

Research is not a separate brand. It is a 6Data mode.

Research project structure

Each research project should include:

```
title
research question
hypothesis
datasets used
dataset versions
cleaning steps
queries
charts
3D snapshots
renders
methods
findings
limitations
conclusion
references/notes
export
```

Research editor sections

```
Overview
Question
Datasets
Methodology
Analysis
Charts
3D Views
Findings
Limitations
Conclusion
Export
```

Reproducibility log

Auto-generate:

```
Raw dataset imported at X time
Dataset cleaned using Y steps
Query Z produced chart A
Dashboard/report used dataset version B
Report exported at C time
```

Research exports

MVP:

```
Markdown
```

Later:

```
PDF
HTML share page
presentation format
```

Research user journey

```
Create research project
Attach dataset
Clean dataset
Run query
Create chart
Create 3D view
```

```
Write methods
Write findings
Add limitations
Export report
```

---

18. Learning Mode Inside 6Data

Goal

Teach users through their own data.

Learning is not a separate app. It is a 6Data mode.

Learning topics

```
SQL basics
SELECT / WHERE / ORDER BY
GROUP BY
aggregations
joins
data types
missing values
duplicates
data cleaning
chart selection
mean / median / mode
variance
correlation
outliers
normalisation
database relationships
basic research methods
```

Dataset-aware learning

If a user has a dataset, generate lessons around it.

Examples:

```
Your dataset has a date column. Learn how to group by month.
Your dataset has categories. Learn how to count by group.
Your dataset has missing values. Learn how to inspect nulls.
Your dataset has numeric columns. Learn average, median and outliers.
```

Learning exercise format

```
Concept
Example
Task
Hint
Editor / controls
Run answer
Result
Explanation
Next challenge
```

SQL exercise example

```
Task:
Find the top 10 artists by total listening time.

Expected concept:
GROUP BY and SUM.

User writes SQL.
App runs it.
App explains the result.
```

Learning levels

```
Beginner
Intermediate
Advanced
Research mode
```

---

19. Automation Centre

Goal

Make datasets and dashboards living systems instead of static uploads.

Automation types

MVP:

```
manual refresh
run cleaning pipeline after import
quality warning after import
refresh dashboard manually
```

Later:

```
scheduled import
scheduled dashboard refresh
alert when value changes
alert on missing data
detect anomalies
export report on schedule
sync connector daily
```

Automation model

```
trigger
condition
action
history
status
```

Examples:

```
When CSV is imported, run this cleaning pipeline.
When dataset quality score drops below 70, show warning.
When dashboard refreshes, update charts.
When connector syncs, profile new data.
```

---

20. Publishing Hub

Goal

Let users share and export work.

Export types

MVP:

```
CSV export
JSON export
```

```
Markdown report
PNG chart
PNG 3D snapshot
PNG render
```

Later:

```
PDF report
public dashboard
private share link
HTML report
video render
interactive embed
```

Privacy modes

```
Private
Workspace only
Anyone with link
Public
```

Default should always be **Private**.

Public/share pages later

```
/share/dashboard/[id]
/share/report/[id]
/share/render/[id]
```

---

21. Connector Marketplace

Goal

Let data enter 6Data from multiple sources over time.

This is a marketplace of connectors, not separate product brands.

Connector categories

```
Files
Databases
APIs
```

```
Apps
Manual
Demo
```

Initial connectors

```
CSV
JSON
Manual Table
Demo Dataset
6Stats Demo Import
```

Later connectors

```
Excel
SQLite
Postgres
Supabase
REST API
Google Sheets
6Stats real account connection
```

6Stats connector

This is the most important ecosystem connector.

6Stats data types:

```
listening events
tracks
artists
daily stats
top tracks
top artists
listening history
audio features
```

6Data should be able to use 6Stats data for:

```
music analytics dashboard
3D music taste cloud
Audio DNA render
listening heatmap
```

```
research project
learning exercises
```

---

22. 6Play Direction

6Play should remain the music/audio software direction.

Do not build it inside 6Data.

But 6Data can later analyse data produced by 6Play.

Possible future relationship:

```
6Play creates/plays/manages music/audio.
6Stats analyses Spotify/listening behaviour.
6Data imports music/audio/listening data for deeper analysis and research.
```

6Play ideas:

```
desktop music player
audio library manager
playlist tool
audio metadata explorer
music visualiser
creative listening interface
```

Keep 6Play separate from 6Data.

---

23. Database Model Draft

Core tables/entities:

```
users
workspaces
datasets
dataset_versions
data_sources
data_files
schema_columns
profile_reports
cleaning_steps
saved_queries
query_runs
```

```
charts
three_d_views
renders
dashboards
dashboard_widgets
research_projects
research_assets
learning_lessons
learning_attempts
automations
automation_runs
exports
activity_logs
connectors
```

users

```
id
email
display_name
image_url
created_at
updated_at
```

workspaces

```
id
user_id
name
description
visibility
created_at
updated_at
```

datasets

```
id
workspace_id
name
description
source_type
status
row_count
column_count
quality_score
active_version_id
```

```
created_at
updated_at
```

dataset_versions

```
id
dataset_id
version_number
label
raw_file_key
processed_file_key
row_count
column_count
created_at
created_by
```

data_sources

```
id
workspace_id
type
name
config_json
status
last_sync_at
created_at
updated_at
```

schema_columns

```
id
dataset_id
version_id
name
detected_type
final_type
nullable
missing_count
unique_count
sample_values_json
stats_json
created_at
```

profile_reports

```
id
dataset_id
version_id
summary_json
quality_json
warnings_json
suggestions_json
generated_at
```

cleaning_steps

```
id
dataset_id
version_id
order_index
step_type
description
config_json
created_at
```

saved_queries

```
id
workspace_id
dataset_id
name
sql
query_type
created_at
updated_at
```

query_runs

```
id
query_id
dataset_id
dataset_version_id
sql
result_preview_json
row_count
execution_ms
created_at
```

charts

```
id
workspace_id
dataset_id
query_id
name
chart_type
config_json
created_at
updated_at
```

three_d_views

```
id
workspace_id
dataset_id
query_id
name
view_type
mapping_json
camera_json
filters_json
created_at
updated_at
```

renders

```
id
workspace_id
dataset_id
three_d_view_id
name
template_type
mapping_json
style_json
output_file_key
created_at
```

dashboards

```
id
workspace_id
name
description
```

```
layout_json
visibility
created_at
updated_at
```

dashboard_widgets

```
id
dashboard_id
widget_type
source_type
source_id
config_json
layout_json
created_at
updated_at
```

research_projects

```
id
workspace_id
name
question
hypothesis
methods_markdown
findings_markdown
limitations_markdown
conclusion_markdown
created_at
updated_at
```

learning_lessons

```
id
workspace_id
dataset_id
title
topic
difficulty
content_json
created_at
```

automations

```
id
workspace_id
name
trigger_type
trigger_config_json
action_type
action_config_json
enabled
created_at
updated_at
```

exports

```
id
workspace_id
entity_type
entity_id
export_type
file_key
created_at
```

activity_logs

```
id
workspace_id
entity_type
entity_id
action
metadata_json
created_at
```

---

24. Suggested File Structure

If building inside 6ixtype first:

```
src/
  app/
    launchpad/
      page.tsx

    6data/
```

```
    page.tsx
    app/
      page.tsx
    workspaces/
      page.tsx
      [id]/
        page.tsx
    import/
      page.tsx
    datasets/
      page.tsx
      [id]/
        page.tsx
    query/
      page.tsx
    visualise/
      page.tsx
    3d/
      page.tsx
    render/
      page.tsx
    dashboards/
      page.tsx
      [id]/
        page.tsx
    research/
      page.tsx
      [id]/
        page.tsx
    learn/
      page.tsx
    connectors/
      page.tsx
    publishing/
      page.tsx

components/
  launchpad/
    AppCard.tsx
    LaunchpadGrid.tsx

  6data/
    layout/
      DataShell.tsx
      DataSidebar.tsx
      DataTopbar.tsx
      WorkspaceSwitcher.tsx

    landing/
      DataHero.tsx
      DataLifecycle.tsx
```

```
    CapabilityGrid.tsx
    UseCases.tsx
    Roadmap.tsx

  overview/
    OverviewCards.tsx
    RecentDatasets.tsx
    RecentWorkspaces.tsx
    DemoDatasetGrid.tsx

  import/
    ImportSourceGrid.tsx
    FileDropzone.tsx
    ImportPreviewTable.tsx
    SchemaReviewPanel.tsx
    ImportProgress.tsx

  dataset/
    DatasetHeader.tsx
    DatasetTabs.tsx
    DatasetOverview.tsx
    DataPreviewTable.tsx
    SchemaTable.tsx
    ProfileReport.tsx
    QualityScore.tsx
    LineageTimeline.tsx

  cleaning/
    CleaningStudio.tsx
    CleaningStepList.tsx
    CleaningActionPicker.tsx
    BeforeAfterPreview.tsx

  query/
    SqlEditor.tsx
    QueryResults.tsx
    VisualQueryBuilder.tsx
    SavedQueries.tsx

  charts/
    ChartBuilder.tsx
    ChartRenderer.tsx
    ChartMappingPanel.tsx
    ChartRecommendations.tsx

  three/
    ThreeSceneCanvas.tsx
    ThreeScatterPlot.tsx
    ThreeBarField.tsx
    ThreeTerrain.tsx
    ThreeTooltip.tsx
```

```
      ThreeControlPanel.tsx
      ColumnMapping3D.tsx

    render/
      RenderStudio.tsx
      RenderTemplatePicker.tsx
      DataOrb.tsx
      CategoryTowers.tsx
      TimelineTunnel.tsx
      AudioDNA.tsx
      RenderExportPanel.tsx

    dashboards/
      DashboardBuilder.tsx
      DashboardGrid.tsx
      DashboardWidget.tsx

    research/
      ResearchEditor.tsx
      DatasetAttachmentPanel.tsx
      ChartEmbedPicker.tsx
      ReproducibilityLog.tsx

    learning/
      LearningHome.tsx
      LessonCard.tsx
      ExerciseRunner.tsx
      HintPanel.tsx

    connectors/
      ConnectorCard.tsx
      ConnectorGrid.tsx

lib/
  6data/
    demo-datasets.ts
    csv.ts
    json.ts
    infer-schema.ts
    profile.ts
    quality-score.ts
    cleaning.ts
    query.ts
    chart-recommendations.ts
    export.ts
    three-utils.ts
    render-utils.ts

data/
  apps.ts
```

```
    6data-capabilities.ts
    6data-roadmap.ts
```

---

25. Stage Plan

Build stage by stage.

Do not build everything at once.

Stage 1 — 6ixtype Launchpad

Goal

Create the official ecosystem page.

Build

```
/launchpad
```

Include only

```
6ixtype
6Stats
6Data
6Play
```

Filters

```
All
Live
Building
Prototype
Music
Data
Platform
```

Acceptance criteria

```
Launchpad page exists.
Only official ecosystem products are shown.
6Data card links to /6data.
6Stats card links to live 6Stats.
```

```
6Play card exists as direction/prototype.
6ixtype card exists as parent brand.
```

Stage 2 — 6Data Landing Page

Goal

Create the public product page for 6Data.

Route

```
/6data
```

Sections

```
Hero
Data lifecycle
Capabilities grid
Import section
Query/visualise section
3D Lab teaser
Render Studio teaser
Research/Learning section
Roadmap
CTA
```

Hero copy

```
6Data
Your data, understood.

Import files, databases and app data. Profile, clean, query, visualise,
research, learn from and render datasets from one workspace.
```

Acceptance criteria

```
6Data landing page looks like a real product.
It explains the full data lifecycle.
It does not undersell the platform with only 4 feature cards.
It clearly includes Research and Learning inside 6Data.
```

Stage 3 — 6Data App Shell

Goal

Create the internal app layout.

Route

```
/6data/app
```

Build

```
sidebar
topbar
workspace selector
search placeholder
import button
empty state
demo dataset cards
recent activity placeholder
```

Acceptance criteria

```
App shell exists.
Sidebar contains all main 6Data zones.
Responsive layout works.
No backend required yet.
```

Stage 4 — Mock Workspaces + Datasets

Goal

Create the product structure with mock data.

Routes

```
/6data/workspaces
/6data/workspaces/[id]
/6data/datasets
/6data/datasets/[id]
```

Mock objects

```
workspace
dataset
schema columns
profile report
cleaning steps
saved query
chart
dashboard
research project
```

Acceptance criteria

```
Can browse workspaces.
Can open workspace.
Can open dataset.
Dataset page has tabs.
All data is mock/static.
```

Stage 5 — CSV Import UI

Goal

Let users import CSV in the browser for preview.

Route

```
/6data/import
```

Build

```
source selector
file dropzone
CSV parser
preview table
schema inference
schema review
confirm import
temporary dataset creation
```

Type inference

Detect:

```
text
integer
decimal
boolean
date
datetime
category
url
email
unknown
```

Acceptance criteria

```
User can upload CSV.
Preview appears.
Columns are detected.
Types are inferred.
User can confirm import.
Temporary dataset is created in UI.
```

Stage 6 — Dataset Profile Engine

Goal

Generate useful dataset understanding.

Build

```
row count
column count
missing values
duplicates
unique values
numeric summaries
text summaries
date ranges
top values
quality score
warnings
suggested questions
suggested charts
```

Acceptance criteria

```
Profile tab shows real computed stats from imported CSV.
Quality score exists.
Column profiles exist.
Warnings exist.
```

Stage 7 — Data Preview Table

Goal

Make viewing data useful.

Build

```
pagination
search
sorting
column filters
column visibility
type badges
missing value highlight
row detail drawer
```

Acceptance criteria

```
Preview tab is usable.
Large-ish datasets do not destroy the UI.
Users can filter and inspect rows.
```

Stage 8 — Cleaning Studio MVP

Goal

Create tracked cleaning steps.

Build actions

```
rename column
change type
remove duplicates
fill missing
remove missing rows
```

```
trim text
replace values
filter rows
create calculated column
```

Acceptance criteria

```
User can add cleaning step.
Step appears in lineage.
Preview updates.
Raw data remains preserved.
```

Stage 9 — Query Lab MVP

Goal

Allow SQL-style analysis.

Build

```
SQL editor
run query
result table
save query
query history
turn result into chart
```

Acceptance criteria

```
User can run basic SQL or SQL-like query on dataset.
Results render.
Query can be saved.
```

Stage 10 — Visualisation Studio MVP

Goal

Create normal charts.

Build chart types

```
number card
bar chart
```

```
line chart
area chart
donut chart
scatter plot
histogram
table
heatmap
```

Acceptance criteria

```
User can select dataset/query result.
User can map columns.
Chart renders.
Chart can be saved.
```

Stage 11 — Dashboard Builder MVP

Goal

Let users save visual outputs into dashboards.

Build

```
create dashboard
add chart
add number card
add table
add markdown note
move widgets
resize widgets
save layout
```

Acceptance criteria

```
Dashboard can be created.
Saved charts can be added.
Layout persists in mock or DB.
```

Stage 12 — Research Workspace MVP

Goal

Create reproducible research workflow inside 6Data.

Build

```
create research project
write research question
write hypothesis
attach datasets
attach charts
attach 3D snapshots later
write methods
write findings
write limitations
export markdown
```

Acceptance criteria

```
Research page exists.
User can create/edit research project.
Dataset/chart attachments work.
Markdown export works.
```

Stage 13 — Learning Mode MVP

Goal

Teach users from their data.

Build

```
learning home
dataset-aware lesson cards
SQL exercises
data cleaning exercises
chart selection exercises
basic statistics explanations
```

Acceptance criteria

```
User can select dataset.
Learning Mode generates/use displays exercises.
User can run a simple exercise.
Explanation appears.
```

Stage 14 — 3D Lab v1

Goal

Create first real 3D visualisation.

Build first

```
3D scatter plot
```

Column mapping

```
X axis
Y axis
Z axis
color
size
label
```

Interactions

```
orbit
pan
zoom
hover tooltip
click row
reset camera
save view
export screenshot
```

Acceptance criteria

```
User can create a 3D scatter from dataset.
User can map columns.
3D scene renders.
Tooltip works.
Screenshot export works.
```

Stage 15 — Render Studio v1

Goal

Create data-driven visual renders.

Build templates

```
Data Orb
Category Towers
Timeline Tunnel
Data Terrain
Audio DNA
```

Acceptance criteria

```
User can choose render template.
User can map columns.
Render preview appears.
PNG export works.
```

Stage 16 — 6Stats Connector / Import

Goal

Connect 6Stats with 6Data.

First version

Use demo/static import.

Later version

Real account connection.

6Stats data model

```
tracks
artists
listening events
daily stats
top tracks
top artists
audio features
```

6Data outputs from 6Stats

```
music analytics dashboard
listening heatmap
3D music taste cloud
Audio DNA render
```

```
research project
learning SQL exercises
```

Acceptance criteria

```
6Data has 6Stats connector card.
Demo 6Stats dataset can be loaded.
Music analytics template works.
3D music visual works.
```

Stage 17 — Database Connectors

Goal

Make 6Data more serious for technical users.

Start with

```
Postgres
Supabase
SQLite upload
```

Features

```
connect read-only
list tables
preview table
inspect schema
import table snapshot
run read-only query
disconnect/delete credentials
```

Acceptance criteria

```
Database connector UI exists.
SQLite upload works first if easiest.
Postgres/Supabase can be added after.
```

Stage 18 — Automation Centre

Goal

Make workflows repeatable.

MVP

```
manual refresh
run cleaning pipeline after import
quality score warning
refresh dashboard
```

Later

```
scheduled import
scheduled sync
threshold alert
anomaly alert
scheduled export
```

Acceptance criteria

```
Automation page exists.
At least one automation flow works.
Automation run history exists.
```

Stage 19 — Publishing Hub

Goal

Let users export/share their work.

MVP exports

```
CSV
JSON
Markdown
PNG chart
PNG 3D screenshot
PNG render
```

Later

```
PDF
public dashboard
private share link
HTML report
video render
```

Acceptance criteria

```
User can export cleaned dataset.
User can export research markdown.
User can export chart/render image.
```

Stage 20 — AI Analyst Later

Goal

Add AI only after deterministic data system exists.

AI features

```
explain dataset
suggest questions
suggest cleaning steps
generate SQL
explain SQL
suggest charts
summarise chart
draft research report
explain limitations
```

AI rules

```
AI must show dataset used.
AI must show generated SQL.
AI must show filters.
AI must show limitations.
AI must not invent results.
AI must not modify data without approval.
```

Acceptance criteria

```
AI can generate SQL.
User approves before running.
AI output includes method and limitations.
```

---

26. MVP Scope

The first public demo should prove this flow:

```
Upload CSV
→ preview data
→ detect schema
→ profile dataset
→ clean data
→ run query
→ create chart
→ save dashboard
→ create research note
→ create 3D scatter
→ export result
```

Must-have for first MVP

```
6ixtype Launchpad
6Data landing page
6Data app shell
workspaces
CSV import
dataset preview
schema detection
profile report
quality score
cleaning steps
query lab
chart builder
dashboard builder
research notes
learning mode basic
3D scatter
export PNG/CSV/Markdown
```

Should-have

```
Render Studio basic
6Stats demo dataset
Audio DNA render
lineage timeline
saved queries
saved camera presets
```

Not for first MVP

```
real AI analyst
team collaboration
billing
full database connectors
real-time sync
video renders
advanced automations
public marketplace
```

---

27. Demo Datasets

Include built-in demo datasets.

Demo Dataset 1 — Music / 6Stats

Columns:

```
played_at
track_name
artist_name
album_name
duration_ms
tempo
energy
valence
danceability
genre
play_count
```

Use for:

```
music analytics dashboard
3D music cloud
Audio DNA render
SQL learning
research mode
```

Demo Dataset 2 — Research Survey

Columns:

```
respondent_id
age_group
study_hours
sleep_hours
stress_level
grade_band
commute_time
part_time_job
```

Use for:

```
research workflow
correlation
charts
statistics lessons
```

Demo Dataset 3 — Generic Activity Data

Columns:

```
date
category
activity
duration_minutes
score
notes
```

Use for:

```
personal analytics
charts
dashboard builder
learning mode
```

No separate product brand should be created for any demo dataset.

---

28. Core User Journeys

Journey 1 — Beginner uploads CSV

```
Open 6Data
Create workspace
Upload CSV
Review preview
Confirm schema
See profile
Fix missing values
Create chart
Save dashboard
Export
```

Journey 2 — Learner uses own data

```
Open Learning Mode
Select dataset
Choose SQL basics
Run generated exercise
See explanation
Save progress
Try harder task
```

Journey 3 — Research workflow

```
Create research project
Attach dataset
Clean dataset
Run query
Create charts
Create 3D view
Write methods/findings/limitations
Export markdown report
```

Journey 4 — 6Stats into 6Data

```
Open 6Data
Choose 6Stats demo import
Generate music analytics dataset
```

```
Open music dashboard
Create 3D music taste cloud
Create Audio DNA render
Export image
```

Journey 5 — Technical user

```
Create workspace
Import SQLite/Postgres/Supabase later
Inspect schema
Run SQL
Save query
Build dashboard
Export result
```

---

29. Development Rules For Antigravity

Use these rules while building:

```
Do not create extra product brands.
Do not reference removed ecosystem products.
Keep Research and Learning inside 6Data.
Keep 6Data as the main platform.
Keep 6Stats focused on Spotify/music analytics.
Keep 6Play focused on music/audio software direction.
Keep 6ixtype as parent brand/launchpad/portfolio.
```

Implementation rules:

```
Build stage by stage.
Keep every stage buildable.
Use mock data first where backend is not ready.
Use reusable components.
Use TypeScript strictly.
Do not break existing 6ixtype routes.
Do not overbuild AI before data basics work.
Do not make 3D mandatory for normal analysis.
Make raw data preservation a core principle.
Every chart should know its source.
Every research result should be reproducible.
```

---

30. First Antigravity Task

Start with this exact implementation request:

```
Implement Stage 1, Stage 2 and Stage 3 only.

Context:
The ecosystem must only contain:
- 6ixtype = parent brand / portfolio / launchpad
- 6Stats = Spotify/music analytics
- 6Data = main data platform
- 6Play = music/audio software direction
- Research/Learning tools = inside 6Data

Tasks:
1. Add a /launchpad page to 6ixtype.
2. Create app cards only for 6ixtype, 6Stats, 6Data and 6Play.
3. Add a /6data landing page.
4. Add a /6data/app shell with sidebar, topbar, workspace selector, demo
dataset cards, and empty states.
5. Add mock data files for apps, capabilities, roadmap, demo datasets and
workspaces.
6. Do not add backend yet.
7. Do not create any extra product brands.
8. Ensure npm run build passes.
```

---

31. Second Antigravity Task

```
Implement Stage 4, Stage 5 and Stage 6.

Tasks:
1. Create mock workspace and dataset pages.
2. Add /6data/workspaces and /6data/workspaces/[id].
3. Add /6data/datasets and /6data/datasets/[id].
4. Dataset detail page must have tabs:
   Overview, Preview, Schema, Profile, Clean, Query, Charts, 3D, Research,
Lineage, Export.
5. Add CSV import UI at /6data/import.
6. Parse CSV client-side.
7. Infer basic schema.
8. Generate dataset profile stats.
9. Show quality score and warnings.
10. Keep data in local/mock state for now.
11. Ensure npm run build passes.
```

---

32. Third Antigravity Task

```
Implement Stage 7, Stage 8, Stage 9 and Stage 10.

Tasks:
1. Add Cleaning Studio MVP.
2. Add tracked cleaning steps.
3. Add Query Lab MVP.
4. Add SQL-like query editor or visual query builder if SQL engine is not
ready.
5. Add basic chart builder.
6. Add dashboard builder.
7. Make charts traceable to dataset/query.
8. Ensure all pages use existing 6Data app shell.
9. Ensure npm run build passes.
```

---

33. Fourth Antigravity Task

```
Implement Stage 11, Stage 12, Stage 13 and Stage 14.

Tasks:
1. Add Research Workspace MVP inside 6Data.
2. Add Learning Mode MVP inside 6Data.
3. Add 3D Lab v1 with 3D scatter plot.
4. Add Render Studio v1 with at least one data-driven render template.
5. Enable PNG export for charts/3D/render if possible.
6. Keep everything connected to datasets.
7. Do not introduce separate product brands.
8. Ensure npm run build passes.
```

---

34. Final Long-Term Direction

The long-term product should feel like this:

```
6ixtype is the brand.
6Stats proves the analytics direction.
6Data becomes the main platform.
6Play keeps the music/audio direction alive.
Research and Learning become powerful modes inside 6Data.
```

The killer 6Data demo:

```
A user uploads a dataset.
6Data profiles it.
6Data explains data quality issues.
The user cleans it.
The user queries it.
The user creates charts.
The user explores it in 3D.
The user creates a polished render.
The user builds a dashboard.
The user writes a research report.
The user exports or shares the result.
```

That is the core product.